

Episode 01 — The Mystery of the Magnetic Cat

NGSS Standards: 3-PS2-3, 3-PS2-4
Grade 3 · Ages 8-9 · FREE Resource

Companion Worksheet · Magnets & Magnetic Fields

Pip's Discovery Lab ·
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Name: _____

Date: _____

Grade: _____

What You Will Learn

After watching Episode 01, you will be able to:

1. Explain what a magnetic field is and how it works
2. Name the three ferromagnetic metals (hint: I Need Chocolate!)
3. Describe how opposite poles attract and same poles repel

Section 1

Key Vocabulary — Match each term to its definition

Term		Definition
1. Magnetic Field	→	The invisible pushing-and-pulling zone around every magnet
2. Ferromagnetic	→	A metal that is naturally attracted to magnets (iron, nickel, or cobalt)
3. Conductor	→	A material that allows electricity or magnetic force to pass through it
4. Poles	→	The two ends of a magnet — North (N) and South (S)
5. Repel	→	To push away — what happens when the same magnetic poles face each other
6. Attract	→	To pull toward — what happens when opposite magnetic poles face each other

Section 2

Fill in the Blanks — Complete each sentence

1. Only three metals are naturally magnetic: _____, _____, and _____.
2. Pip's memory trick for the three magnetic metals is: _____
(I = Iron, N = Nickel, C = _____)
3. A magnetic field can pass through _____, paper, plastic, and wood.

4. When Pip brought the fridge magnet close to a steel paper clip, the clip moved BEFORE the magnet touched it. This is because of the magnetic _____.
5. North pole + South pole = they _____ (pull together).
6. North pole + North pole = they _____ (push apart).
7. Whiskers's collar tag was pulled by the magnet because it is made of stainless steel, which contains _____.

Section 3 True or False — Circle the correct answer

Statement	True	False
All metals are magnetic.	☐	☐
A magnetic field can work through the air without touching an object.	☐	☐
A copper penny will stick to a horseshoe magnet.	☐	☐
A steel paper clip will stick to a magnet.	☐	☐
North and South magnetic poles attract each other.	☐	☐
North and North magnetic poles attract each other.	☐	☐
Magnetic fields can pass through your hand to reach iron objects.	☐	☐
Aluminum is one of the three ferromagnetic metals.	☐	☐

Section 4 Sort It! — Write each item in the correct column below

Items to sort: steel nail · copper penny · plastic dinosaur · aluminum soda can · wooden block · steel paper clip · rubber band · iron fridge door

MAGNETIC	NOT MAGNETIC
(contains iron, nickel, or cobalt)	(no iron, nickel, or cobalt)

Section 5 Short Answer — Answer in your own words

1. In your own words, explain what a magnetic field is:

2. Why did the aluminum soda can NOT stick to Pip's magnet?

3. Pip's memory trick is 'I Need Chocolate.' What does each letter stand for?

4. What happens when you put two SAME poles of a magnet together?

5. How did Pip solve the Mystery of the Magnetic Cat?

Section 6

Try It At Home! Magnetic Scavenger Hunt

Try It At Home! Magnetic Scavenger Hunt

You need: Any fridge magnet from your kitchen

Step 1: Walk around your house with the fridge magnet.

Step 2: Touch it to 10 different objects — cutlery, coins, toys, door handles, cans.

Step 3: Record what sticks and what doesn't in the table below.

What happened? Which objects stuck to the magnet? What do they all have in common?

My answer: _____

Object I Tested	Did it STICK?	Made of...?	Magnetic? Y/N
1. _____	Yes / No	_____	Y / N
2. _____	Yes / No	_____	Y / N
3. _____	Yes / No	_____	Y / N
4. _____	Yes / No	_____	Y / N
5. _____	Yes / No	_____	Y / N
6. _____	Yes / No	_____	Y / N
7. _____	Yes / No	_____	Y / N

8. _____	Yes / No	_____	Y / N
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Share your results in the YouTube comments! What was the most surprising object that stuck?

Section 7	Reflection — What did you think?
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My favorite part of this episode was: 	One question I still have:
I would rate this episode: ■ ■ ■ ■ ■ (circle one: 1 2 3 4 5 stars)	I would recommend this to a friend because:

Stay Curious! ✦

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